

Beyond Fairness: Educational Responsibility in LLM-Personalised Learning

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Abstract—LLM-based educational personalisation may implicitly encode socioeconomic bias, shaping explanations, expectations, and learning trajectories. Without socioeconomic status (SES)-aware evaluation, such systems risk reinforcing educational inequality rather than mitigating the digital divide. This discussion paper argues that these risks cannot be addressed through technical fairness metrics alone. Instead, we contend that LLM-driven personalisation in education raises broader questions of educational responsibility, including whose values are embedded in learning systems, how learners are positioned through inferred profiles, and who is accountable for resulting harms. We aim to reframe current debates and outline a research and governance agenda for responsible use of LLMs in education.

Index Terms—Large Language Models, Personalisation, Socioeconomic Status, Educational Equity, Bias

I. INTRODUCTION

Educational inequality remains a persistent and well-documented challenge, with students from lower socioeconomic status (SES) facing systemic disadvantages in access, attainment, and educational outcomes [1]–[3]. In response, recent discourse has positioned Artificial Intelligence in Education (AIED) as a promising mechanism for mitigating these disparities, particularly through personalised and adaptive educational technologies [8]. Large Language Models (LLMs) support this perspective given their capacity to generate human-like, context-aware responses and explanations. However, challenges remain in the evaluation and reduction of SES biases that may shape interactions in ways that reproduce or amplify existing inequalities rather than reduce them.

SES is a multidimensional construct characterising the relative social and economic position of individuals or groups through their access to financial, social, cultural, and human-capital resources [2], which in turn shapes students’ learning experiences and outcomes [3]. In AIED, SES is commonly interpreted through either deficit perspectives (lack, risks, and disadvantages) or asset-based perspectives (strengths, knowledge, and cultural wealth) [5], [6]. Given this multidimensionality, LLMs may implicitly infer SES from linguistic and family background cues. When such characteristics are inferred rather than explicitly modelled, the assumed benefits of personalisation become less straightforward. Unlike earlier search technologies that primarily retrieved information [9], LLMs actively mediate learning by framing explanations and shaping what is presented as relevant or achievable. From a deficit perspective, these inferences may disadvantage marginalised learners by lowering cognitive expectations or simplifying

content, thereby shaping the level of abstraction, pedagogical depth, and examples provided. Consequently, evaluation frameworks should examine how LLMs may reproduce deficit assumptions and whether they can recognise and build on learner assets [5], [6].

LLMs are increasingly integrated in educational systems, yet the educational implications of their biases remain under-explored. Biases in LLMs have been widely documented in social, political, health, legal, and hiring domains, including both representational harms (e.g., stereotyping, exclusionary norms, and disparate performance) and allocational harms (e.g., direct and indirect discrimination) [11], [12]. In education, benchmarking models across diverse socioeconomic contexts is necessary to ensure AIED tools help bridge rather than widen the digital divide [13]. When such biases remain implicit, they may shape learners’ epistemic perspectives, self-perception, and worldviews in ways that systematically disadvantage certain populations [7]. Moreover, the predominance of training data drawn from a narrow set of languages, regions, and high-income cultural contexts raises concerns about whose knowledge and educational values are encoded and whose are marginalised [11]. Furthermore, the high computational and financial costs of LLM deployment may reinforce educational inequalities through subscription models, institutional licensing, and unequal infrastructure [10].

II. SES BIAS AND PERSONALISING WITH LLMs

LLM-driven personalised education suggests that tailoring responses based on user profiles improves relevance, clarity, and engagement. However, this may introduce both intrinsic bias (inherent in the model’s embeddings, where latent representations encode socioeconomic patterns learned from training data) and extrinsic bias (affecting the fairness of system outputs in downstream tasks). Consequently, interactions that vary in linguistic form or implied background can systematically trigger distinct internal pathways, leading to responses that reflect socioeconomic stratification rather than only intentional pedagogical adaptation.

Representation activation. LLMs extract socioeconomic cues from minimal linguistic signals such as names, occupations, or framing choices and associate them with wealth, poverty, or social status even in the absence of explicit labels [14]. In personalised educational settings, prompts that encode learner background (e.g., interests, experiences, or prior knowledge) may inadvertently activate latent SES-related

dimensions within the model. This means that two learners asking the same question with differently framed prompts can receive responses that diverge not only in tone or complexity, but also in the assumptions embedded about social position.

SES-sensitive interaction dynamics limit learning personalisation. A recent survey of real-world LLM usage shows systematic differences in how individuals from different socioeconomic backgrounds in the US and UK formulate prompts, including variation in abstraction, verbosity, politeness, and task framing. Because LLMs are highly responsive to such linguistic features, these differences propagate to model outputs, reinforcing an interaction loop in which higher-SES prompt styles elicit more concise, abstract, or authoritative responses, while lower SES styles risk eliciting over-simplified, generic, or constrained outputs [13]. Considering that the US and the UK already score above the PISA 2022 mean in reading literacy, countries below the mean may face a greater risk of AI becoming a rigid cognitive ceiling that privileges high-SES linguistic norms and limits complex reasoning.

Personalising Educational Responses with LLMs suggests that the effectiveness of specific prompting strategies is deeply tied to the learner’s existing background (knowledge, interests, and preferences). For term explanations, knowledge-level personalisation is the most effective, but its benefits depended on students’ prior background: it provided little advantage over hobby-based personalisation for students with weak foundations, while consistently outperforming other strategies for students with stronger backgrounds [8]. When LLMs function as teachers, they exhibit significant biases in ranking and generation across demographic groups, including race and disability status [4]. This indicates that LLM response framing leverages the cultural and educational capital provided in a user profile to construct meaning. Because profile attributes such as hobby access or prior technical experience are often stratified by socioeconomic status, personalisation may inadvertently generate uneven educational support that reproduces existing social inequities.

III. CHALLENGES AND CONSIDERATIONS

We draw attention to two interrelated challenges that remain insufficiently addressed despite advances in detection, auditing, and mitigation: First, auditing power and responsibility dynamics requires moving beyond reactive bias correction toward a structural examination of who controls data, whose values are embedded in model design, who benefits from deployment, and who bears responsibility when harms occur. Addressing bias solely at the output level risks obscuring deeper governance issues, particularly when equity considerations are subordinated to efficiency/scalability. Robust governance frameworks are therefore needed to ensure accountability across the full lifecycle of LLMs and to expand fairness analyses beyond English-dominant, Western-centric contexts.

Second, conceptualizing, mitigating, and managing bias, including implicit and socioeconomic biases, remains a fundamental challenge. Such biases are difficult to precisely define, measure, and address, requiring continuously refined

conceptual frameworks and evaluation criteria. Effective bias assessment must move beyond mitigation alone to include management strategies that empower end-users, explicitly characterize performance fairness trade-offs, and clearly specify who is harmed and how. This requires analysis across multiple levels of the model, from latent representations, where issues of encoding and interpretability arise, to observable outputs. However, such multi-level analyses are computationally demanding and remain largely inaccessible at scale.

IV. CONCLUSION

Personalisation should not be conflated with fairness. Bias in inferred socioeconomic status poses a fundamental challenge for AI-driven education, as it risks embedding existing social hierarchies into learning systems. Achieving educational equity therefore requires moving beyond reactive fixes toward inclusive frameworks that prevent such reinforcement by design. As a discussion paper, this work aims to shift the conversation from whether LLMs are biased to how responsibility for their educational use is defined, distributed, and enforced. We argue that without explicit attention to governance, accountability, and pedagogical values, LLM-driven personalisation risks undermining education’s role as a public and democratic good.

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